

Math 534H - Spring 2026 - Lecture 22

①

Heat / Diffusion Eq. with zero Neumann boundary conditions and initial condition:

$$\begin{array}{l} \text{Heat} \\ \text{BIVP} \end{array} \quad \begin{array}{l} \underline{\text{EQ}} \\ \underline{\text{BC}} \\ \underline{\text{IC}} \end{array} \left\{ \begin{array}{l} u_t - k u_{xx} = 0 \quad 0 < x < L, t > 0 \\ u_x(0, t) = 0 = u_x(L, t) \quad \forall t > 0 \\ u(x, 0) = \phi(x) \quad x \in [0, L] \end{array} \right.$$

Same as before we look for solutions of $\underline{\text{EQ}} + \underline{\text{BC}}$ of the form:

$$u(x, t) = X(x) \cdot T(t) \quad \left(\begin{array}{l} X(x) \neq 0 \\ T(t) \neq 0 \end{array} \right)$$

Plug into the EQ. and analyze. to obtain

(2)

$$\begin{cases} \text{2nd order ODE} & X''(x) = -\lambda X(x) \\ \text{1st order ODE} & T'(t) = -\lambda k T(t) \end{cases}$$

Assume $\lambda > 0$ first $\lambda = \beta^2$, $\beta > 0$ and
get then that

$$X(x) = C \cos(\beta x) + D \sin(\beta x) \text{ and}$$

the zero Neumann conditions tell us that

$$X'(0) = 0 = X'(L).$$

$$X'(x) = -\beta C \sin(\beta x) + \beta D \cos(\beta x)$$

Since $\beta > 0$ (nonzero) $X'(0) \Rightarrow D = 0$

$$\text{and } X'(L) = 0 = -\beta \cdot C \sin(\beta L)$$

$\beta > 0$ and $C \neq 0$ because otherwise $X(x) \equiv 0$ for $x \in (0, L]$

$$\text{Hence (as before) } \sin(\beta L) = 0$$

$$\Leftrightarrow \beta L = n\pi \quad (\Rightarrow) \quad \beta = \frac{n\pi}{L} \quad n \geq 1.$$

Exactly as before for $\lambda > 0$ $\lambda = \beta^2 = \left(\frac{n\pi}{L}\right)^2$

What is $T(t)$ in this case?

$$T(t) = A \cdot e^{-\left(\frac{n\pi}{L}\right)^2 \cdot k \cdot t} \quad n \geq 1.$$

So all in all for $\lambda > 0$, $n \geq 1$ we get (4)

$$u(x, t) = X(x) \cdot T(t)$$

$$u(x, t) = C_n \cos\left(\frac{n\pi}{L} x\right) \cdot A_n e^{-\left(\frac{n\pi}{L}\right)^2 kt}$$

$$u_n(x, t) = \tilde{A}_n \cos\left(\frac{n\pi}{L} x\right) e^{-\left(\frac{n\pi}{L}\right)^2 kt} \quad \text{where}$$

$$C_n \cdot A_n = \tilde{A}_n$$

Now just as in the wave case with zero Neumann

BC. we note that $n=0$ doesn't give us

a trivial solution! In fact $\lambda=0 \Leftrightarrow n=0$

is possible. When $n=0$ we get

(5)

$X(x) = C$ ($C \neq 0$) is a solution

and $T(t) = A$ ($A \neq 0$) is a solution

So when $n=0$ $u(x,t) = \text{constant} \neq 0$

is a possible solution. For convenience let's

call this constant $\frac{\tilde{A}_0}{2}$. All in all we have

$$u_N(x,t) = \frac{\tilde{A}_0}{2} + \sum_{n=1}^N \tilde{A}_n \cos\left(\frac{n\pi}{L}x\right) \cdot e^{-\left(\frac{n\pi}{L}\right)^2 k \cdot t}$$

is a solution to Heat EQ + zero Neumann BC.

We still need to match the initial condition ⑥

$z(x, 0) = \phi(x)$. We need to know if

$$\phi(x) \stackrel{?}{=} \frac{\tilde{A}_0}{2} + \sum_{n=1}^N \tilde{A}_n \cos\left(\frac{n\pi}{L} x\right) \quad \textcircled{+}$$

(note this is the same cosine sum/series as for the initial position in the wave eq. also with zero Neumann conditions)

The answer to ? in $\textcircled{+}$ will follow from the study of Fourier series that we'll do next time

To finish today let's consider a ^{zero} mixed boundary condition. For example suppose you have

$$\begin{cases} u_t - k u_{xx} = 0 & 0 < x < L, t \geq 0 \\ u(0, t) = 0 = u_x(L, t) \quad \forall t \\ u(x, 0) = \phi(x) \end{cases}$$

$u_x(0, t) = 0 = u(L, t)$
Neumann Dirichlet

We are putting zero Dirichlet at $x=0$ but zero Neumann at $x=L$. For $X(x)$ we

get DN

$$\begin{cases} X''(x) = -\lambda X(x) \\ X(0) = 0 = X'(L) \end{cases}$$

We could have also do the reverse namely

⑧

ND

$$\begin{cases} X''(x) = -\lambda X(x) \\ X'(0) = 0 = X(L) \end{cases}$$

In DN case you'll find that $\lambda > 0$ is
only case where we have nontrivial solutions

In ND case you'll find that $\lambda \geq 0$
is possible for nontrivial solutions.

(9)

In either case you'll find that

$$\beta = \frac{(n + \frac{1}{2})\pi}{L}$$

$$\lambda_n = \beta_n^2$$

$$n \geq 1$$

or

$$n \geq 0$$

and:

For **DN** $X_n(x) = C_n \sin\left(\frac{(n + \frac{1}{2})\pi}{L} x\right)$ $n \geq 1$

For **ND** $X_n(x) = A_n \cos\left(\frac{(n + \frac{1}{2})\pi}{L} x\right)$ $n \geq 0$

where C_n and A_n are constants.

The equations for $T(t)$ don't involve the BC. so they are the same as before.